

EEG Quality Analysis — Summary & Validation

1) What Quality Analysis does (short & clear)

Quality Analysis evaluates the reliability of EEG recordings by computing channel-wise metrics such as signal-to-noise ratio (SNR), variance, kurtosis (spikes/outliers), skewness, and line-noise contamination. It does not modify the data; it reports whether the data are suitable for downstream analysis.

2) Why Quality Analysis is useful in EEG

EEG is highly sensitive to artifacts (movement, muscle activity, power-line noise). Running Quality Analysis before feature extraction or classification prevents garbage-in/garbage-out, helps identify bad channels, and guides targeted preprocessing.

3) Quality metrics summary (per channel)

Channel	Score	SNR (dB)	Variance	Kurtosis	Skewness	Line Noise Ratio
channel_1	70	9.2	19.57	-0.84	-0.01	0.002
channel_2	70	8.7	18.63	-0.91	0.08	0.002
channel_3	45	0.9	40.65	73.50	3.39	0.008
channel_4	70	8.5	19.90	-0.78	-0.08	0.002
channel_5	70	8.2	19.34	-0.81	-0.02	0.002
channel_6	70	12.7	45.67	-0.50	0.04	0.000
channel_7	70	8.6	19.18	-0.52	-0.04	0.001
channel_8	70	9.8	18.82	-0.69	-0.11	0.001

Figure 1 — Channel Quality Scores

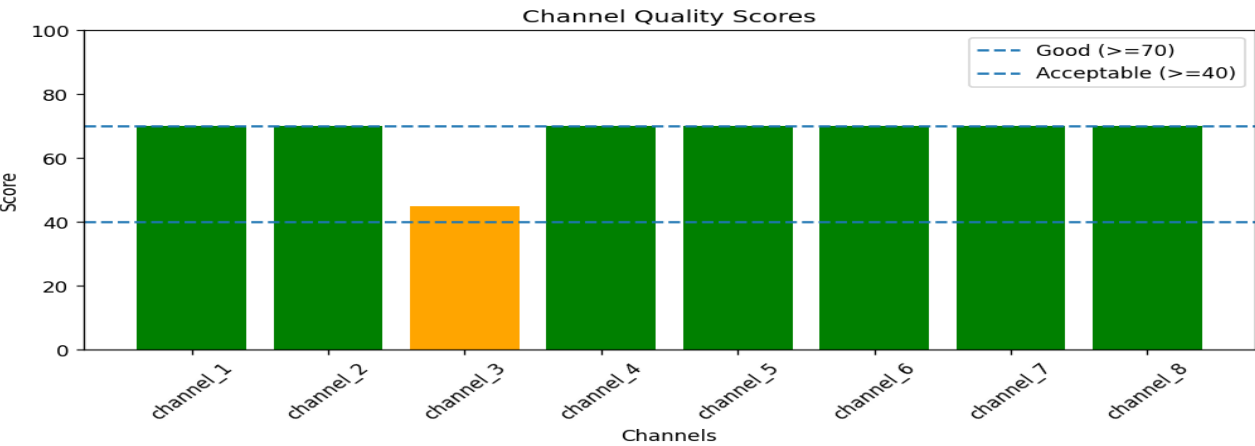
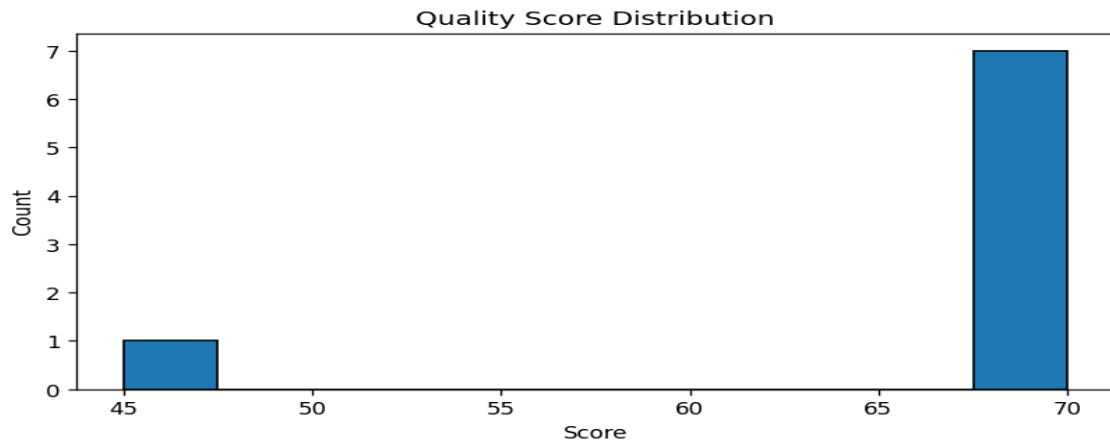


Figure 2 — Quality Score Distribution



Key takeaway

The dataset shows an average quality score of 66.9/100. Channels with high kurtosis/skewness or elevated line-noise ratios receive lower scores, which indicates potential spikes/outliers or electrical contamination. Use these results to exclude or clean poor channels before feature extraction.